

Intaliq Project Milestone 3

SE365: Human Computer Interaction

Section: 1350

Instructor

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Table of contents

Table of Contents	2
1. Title and Introduction	3
2. High-Fidelity Prototype Overview.....	3
2.1. Prototype Description.....	3
2.2. Prototype Materials Used.....	4
2.3. Design Improvements.....	5
3. DECIDE Framework.....	5
4. High-Fidelity Evaluation.....	7
4.1. User Evaluation	8
4.1.1. Evaluation Setting.....	8
4.1.2. Responses.....	9
4.2. Coach Evaluation.....	10
4.2.1. Evaluation Setting.....	10
4.2.2. Responses.....	10
4.3. Expert Evaluation.....	11
4.3.1. Evaluation Setting.....	11
4.3.2. Responses.....	12
4.4. Findings.....	15
4.4.1. User Evaluation Findings.....	15
4.4.2. Coach Evaluation Findings.....	16
4.4.3. Expert Evaluation Findings.....	16
4.5. Discussion.....	17
4.6. Conclusion.....	18
5. ProtoType Screenshots.....	18
6. References.....	19
7. Appendices.....	19
8. Individual Contribution	20

1. Title and Introduction

Saudi Arabia's Vision 2030 places strong emphasis on improving citizens' health and increasing participation in physical activities. Despite this, many individuals still face challenges when engaging in outdoor activities such as running, cycling, and hiking. These challenges include difficulty finding suitable partners, lack of motivation, and limited access to organized, professionally led group sessions. Existing fitness applications mainly focus on individual tracking, while other platforms support activity discovery but lack community based and guided experiences.

To address these limitations, *Intaliq* was developed as an outdoor fitness application that navigates activity tracking, goal setting, and community engagement into a single platform. The system enables users to monitor their progress, join group sessions, connect with fitness partners, and interact with coaches, while also providing AI supported recommendations to enhance overall experience.

Following the development of the low-fidelity prototype in the previous milestone, this milestone focuses on the design and evaluation of a high-fidelity prototype. The improved prototype incorporates feedback obtained from earlier evaluations and aims to provide a more realistic and interactive user experience. Both user and expert evaluations were conducted to assess the system's usability, functionality, and overall user experience, ensuring that the application meets its intended goals and aligns with the usability principles.

2. High-Fidelity Prototype Overview

2.1. Prototype Description

Intaliq's high-fidelity prototype is a multi-functional application that offers many important features for both users and coaches. After creating an account and filling out relevant information, regardless of user type, *Intaliq* users are sent a 6 digit code for verification through their email address.

User prototype:

Once they are let in, users are greeted immediately with a dashboard that has a summary of distance covered, calories burned, active days, and sessions joined. They are also presented with the option to select from the quick actions provided such as, logging an activity, setting a goal, finding sessions, and chatting with an AI Coach. Since session information can change over time, they have a tab titled "Important Announcements" related to any updates in sessions they joined. In regards to the 6 tabs at the bottom of the screen, users have a home, events, activities, sessions, goals, and more tab. Events is the notification center that is categorized by activity type, and there's also a category for general announcements titled "All". Besides that, users can log activities and set goals in

the respective tabs. The sessions tab displays all sessions offered and relevant data, this is where users can request to join sessions. The more tab is where more personal features appear. Viewing profile is where users can change anything about their profile details. Users can connect with other users or even coaches by sending a request and messaging them. Our AI chatbox feature helps in fitness-related advice while holding no medical responsibility. To ensure all those living in Saudi Arabia are able to use our app, we added a language feature that will be more efficiently implemented in future versions. To make sure our users are most comfortable, we have a support email listed for any queries or concerns. Lastly, users can log out whenever they need to.

Coach prototype:

After creating an account/signing in, coaches are presented with a coach dashboard that summarizes their currently active sessions, total members in those sessions, pending user requests, and total sessions in general. Their quick actions include creating a new session, reviewing requests, making announcements, or also viewing their sessions. They have an area in their dashboard that shows their upcoming sessions as well. When it comes to their tabs at the bottom, they have home, events, sessions, admissions, announcements, and more tabs. Events is the notification center that is categorized by activity type, and there's also a category for general announcements titled "All". Sessions displays the coaches own sessions, but also other coaches sessions. They are able to create sessions in this tab as well. In the admissions tab, they are divided by each session and the user who requests to join them. The announce tab is to make announcements that users can view, and if the coach made a general announcement it will show up in the all category of the events page. Coaches more tab functions exactly like a users more page. Offering them to edit their profiles, find partners, chat with AI, change language (future versions), contact support team, or simply log out.

2.2. Prototype Materials Used

To achieve a working prototype, we utilized multiple platforms. For the actual graphical user interface, we worked on Figma to create the designs inspired by the low-fidelity prototype in milestone 2 with the needed adjustments. After that, we utilized VS Code's development environment which was mainly used to edit the files, run commands in the terminal, and most importantly manage project changes. The frontend build tool that we used is Vite. It was used to run the app during development locally, as well as creating optimized production files for deployment. JavaScript, CSS, and HTML were all helpful when it came to our prototype interface, navigation, different dashboards, profile pages, and styling. We have prior knowledge in coding with Java, however these languages were still new to us in terms of syntax and logic. That's why we had to use the help of Codex for any codes we weren't sure about, or needed help with in any way. As a backend service, from our prior knowledge of databases through the CS340 course, we chose Supabase for creating accounts, user authentications (6 digit code), and overall data

storage. Supabase and its features were a big help in bringing our prototype to life, we also used the Supabase Authentication, as well as the Supabase SQL Editor. Our coaches and users are able to create accounts using their emails, which receive a 6 digit code they are asked to put in before continuing, that was done with the help of the previously mentioned Supabase Authentication. For the database schemas to be created, updated, and to follow constraints assigned to any attributes, Supabase's SQL Editor came through with that. The heavy load of actually deploying our application was dealt with using Vercel which allowed our prototype to be accessed using a public link (found in appendices). Github was used to store the project's code and helps in viewing and tracking changes.

2.3. Design Improvements

The difference between our low and high fidelity prototype is major. We added and modified the layout and features to make the app more organized and consistent. In the low fidelity prototype users did not understand the idea of where the notifications go. That is why we improved the high fidelity prototype and added the events tab where it shows notification for running, hiking, biking, and all. We also added when booking a group session if the session is accessible for wheelchair users. In the low fidelity prototype the more tab only had finding partners and AI chatbox. On the other hand, the high fidelity prototype has "view profile", "Language", "help", "logout", "find partners" and the ai chatbox. We also added the ability to search for friends by typing their name.

3. DECIDE Framework

Determine the Goals:

The ultimate goal is to ensure user experience and usability.

Usability:

- Safety when using and protecting users from undesirable conditions
- Easy to learn
- Easy to remember how to use

User experience:

- **Intaliq** is seen as a sleek and modern fitness application
- All users view it as simple, distinct, and pleasurable
- All users view **Intaliq** as exciting, motivating, and rewarding
- Finding a community within your area that shares your interests could be seen as more pleasurable

Explore the Questions

Our questions aim to assess our usability and user experience goals.

Coach:

1. Will coaches be excited to open up *Intaliq* on a daily basis as to input their fitness data?
2. Will coaches be able to easily navigate through our app?
3. After a while of not using, will coaches remember how to use the app?
4. Will the application's colors/aesthetic be eye-catching to coaches?
5. Will coaches find the administrative tasks they usually do more motivating and rewarding through our app?
6. After successfully hosting a fitness session, will coaches feel more rewarded and therefore eager to continue?
7. Will expanding the fitness communities in Saudi Arabia cause an overall increase in physical engagement?
8. Will being able to directly chat with users allow room for more improvement on both ends, specifically in motivating and keeping track of users progress?
9. Will being able to directly chat with other coaches allow room for coaching method insight and overall increasing sessions' quality?
10. Will being able to update and announce to users about upcoming events help in spreading the word about these sessions effectively?

User:

1. Will users be excited to open up *Intaliq* on a daily basis as to input their fitness data?
2. Will users be able to easily navigate through our app?
3. After a while of not using, will users remember how to use the app?
4. Will the application's colors/aesthetic be eye-catching to users?
5. After hitting fitness goals, will users feel more rewarded and therefore eager to continue?
6. Will users find it more pleasurable to find a community that shares their interest that's also within their area?
7. Will having colorful icons with meaningful logos make it more entertaining and rewarding for users?
8. Will the categorization of fitness data help users feel a higher sense of achievement?
9. Will being able to chat directly with coaches help with motivation and accountability?
10. Will being alerted on upcoming events be helpful in planning out fitness activities?

Choose the Evaluation Method

The method used for the evaluation was an interview.

Identify the Practical issues

Finding certified coach user type for the interview took a while due to the limited number of coaches we were able to contact. Especially coaches who were experts in the sports we focused on, including hiking, biking, and running.

Milestone 2 was by far the hardest to execute since it was during our Ramadan break, the exhaustion from fasting all day was definitely an obstacle we encountered.

Staying on schedule, specifically as students with more than one project, was definitely a challenge.

Decide how to deal with the Ethical issues

We made it clear in the form's content that they had the freedom to leave whenever they felt they needed to. Ensuring that our users knew their rights by making them sign our *Intaliq* consent form which included the aim of our project, and what our findings from their contribution will be used for. Highlighting our integrity, users' information was kept private at all times. In the interview process, we also briefly mentioned our project's end-goals.

Evaluate, analyze, interpret and present the data

As mentioned before, we have chosen the interview as our evaluation method.

Ecological validity, in our case, is one of the most important aspects to be considered. When mentioning ecological validity, we raise the question, is the environment influencing the findings? We believe that our environment definitely does. One of our main practical issues was actually finding a Saudi Arabian resident who is also a certified coach. Even mentioned by one of our coach interviewees, Coach Yousif emphasized the difficulty to find coaches in Saudi. He passionately explained how you would only know coaches if you actively went to a gym. In Saudi, fitness communities are there, however they are extremely rare or exclusive. That's why having a digital community that can also be a real-life community is so beneficial in this society. As Coach Yousif explained, the only way to access coaches services is by knowing one already who you can ask to tell you if there are any upcoming events/sessions. This point can be further supported by the fact that Saudi Arabia Vision 2030 has a whole pillar dedicated to achieving a vibrant society. In fact, Saudi has already made progress by raising the life expectancy from 77 years to 78.1 years. Furthermore, there are still objectives in the nation to increase public participation in sports & athletic activities, meaning that these goals are still in the works. With an application like *Intaliq*, the hard parts of physical activity participation seem almost transparent. Our purpose is to solve this environmental obstacle, by making it accessible to those even far from major capital cities.

The evaluation followed a structured process adapted from the DECIDE framework.

4. High-Fidelity Evaluation

All evaluations followed the same procedure, and it went as follows:

Pre	- Ice breakers - Brief review of the project and purpose - Explain that the prototype is high-fidelity
During	- Walk participants through the high-fidelity prototype screen by screen - Briefly go over other user type's prototype

	- Evaluate the interface using a heuristic evaluation checklist based on usability principles - Identify usability issues based on usability principles
Post	- Thank participant for participation - Summarize key observations - Compare feedback with other group members - Identify any usability issues - Suggest improvements

Test plan evaluation justification:

The test plan evaluation method was selected due to our course's emphasis on Nielsen's heuristics. Through his extensive research, he was able to settle on 10 important rules that systems should follow to achieve usability and end up with a user experience. We created a table that mentioned the heuristics with features that align with them accordingly. Allowing all those who will evaluate to check each point on its own.

4.1. User Evaluation User Test Plan.pdf

4.1.1. User Evaluation Settings

Member	Participant	Duration	Mode
Danah AlOmayrini	Sultanah AlOmayrini	2:48 minutes	In person
Ghala Alsaqer	Lamar Alsaqer	1:25 minutes	In person
Sara Alsharif	Majed Alsharif	6 minutes	In person

Average evaluation duration: 3.6 minutes

4.1.2. User responses

Sultanah' response: Sultanah AlOmayrini User Test Plan

When it comes to the overall experience, Sultanah mentioned that she liked the application and described it as simple, clean, and not overwhelming. She stated that she was able to understand most features quickly and felt it is something she would actually use. She also emphasized that the design being simple made it more functional and easier to interact with.

Regarding navigation, Sutlanah confirmed that it was very easy to move between pages, especially when using the bottom menu bar. she did not experience any confusion while navigating through the application. When asked if anything was unclear, she mentioned that everything was generally clear, but suggested adding examples or prompts in the AI type box to help users understand what kind of questions they can ask.

In terms of what she liked the most, sultana highlighted the simplicity of the design and the overall views. She appreciated how the interface focuses on functionality without being complicated. Additionally, she mentioned that the application could motivate her to be more active, especially because of features like joining sessions and tracking progress toward the goals. She explained that seeing progress visually makes her feel encouraged to continue and teach her goals.

Majed's response: Majed Alsharif (User) Test Plan

Overall, Majed believes the *Intaliq* prototype is off to a great start. He claims that the user interface (UI) is clear, polished, and simple to use, with clear icons that make the experience seem natural and require little explanation. He notes that navigating the app is simple and predictable due to the layout's obvious tab organization and constant placement of features like the back button. Additionally, he notes that the bottom navigation approach is robust and complements contemporary app design, facilitating users' rapid access to various sections. Majed emphasizes that, once completely implemented, the usage of pop-up messages for operations like deleting sessions, submitting or canceling requests, and managing full sessions is a clever feature that will boost user confidence. He observes that the error password feedback during login functions effectively and offers precise instructions without creating dissatisfaction. Additionally, he says the regular use of the logo and color palette provides the app a polished, upscale appearance while maintaining the content's relevance and simplicity. He notes that while several sophisticated features, such as the AI chatbox and various pop-up interactions, are not yet fully functional, the concepts underlying them are sound and obviously user-focused. He is impressed with the direction it is taking and believes it is a well-considered prototype that shows a good grasp of usability and design.

Lamar's response: Lamar Alsaqer's User Test Plan

Lamar's overall evaluation of the *Intaliq* app was mixed, with several issues identified. The bottom menu bar icons were functional and easy to understand.

Icons communicated navigation and real world objects without needing text, the layout was clear for all user types, and the color scheme was consistent throughout. Login error handling also worked, displaying an "invalid login credentials" message when entering invalid credentials. However, there were a few significant problems: pop-up messages for actions like removing a session or canceling a booking did not exist, the back button that is supposed to be at the top right of all screens was missing on most screens. Furthermore, the AI chatbox and help button existed but did not function at all. Overall, while the app's visual design and navigation structure were solid, the missing pop-ups, inconsistent back button, and non-functional features need immediate attention.

4.2. Coach Evaluation Coach Test Plan.pdf

4.2.1 Coach Evaluation Settings

Member	Participants	Duration	Mode
Yara Albugami	Coach Yousif	44 minutes	Online
Haya AlJadaan	Dr. Oluwatoyin Toriola	23 minutes	No interview was conducted due to Professor's tight schedule.

Average coach evaluation time: 33 minutes and 30 seconds

4.2.2 Coach responses

Coach Yousif Response: Coach Yousif Test Plan

When it comes to our pop up messages, it seems that coach Yousif faced an issue in viewing them. It was mentioned to him that not all pop up messages will take up the entire screen as he assumed. He did see them, however, they weren't what he expected as we made them a smaller size. He stated that our bottom menu bar was functional. Coach Yousif said that the back button was missing in some pages which he found a bit frustrating to deal with. On the other hand he saw that the logout feature worked and it showed a confirmation pop up message. Since our database attempts to be as accurate as possible, we didn't make pre-made users to automatically join his sessions to test if the application admits users into already full sessions. This is why he believes that feature doesn't work. When he attempted to create sessions on an invalid date that expired he did receive a pop up message that showed this was an error. About icons illustration, he felt that the

events icon has room for improvement and can be better. From his trial run, he experienced a layout structure that was overall good and clear. Coach Yousif noticed that the neon green color has 2 variants and that the information presented was relevant. When he tried to log in with an incorrect password, it gave him a pop up message clearly indicating the error and instructing him on the correct actions to take. He saw that the AI Chatbox and help button were not functional. Just to set the record straight, the help button was only to show the email users and coaches are expected to message for any support needed.

Dr. Oluwatoyin Response: **Dr. Oluwatoyin Coach Test Plan**

Dr. Oluwatoyin found popups unresponsive, with issues regarding the features themselves. However, the popup that appeared when creating a session on an invalid date had worked and provided a clear reason. Also, when logging in with an invalid password, she reported a popup message that told her credentials were invalid. To her, the bottom menu bar functioned clearly and the icons represented the features as they are and took her to the desired pages. Back button and log out were both available and worked as intended for the Dr., and so did the menu bar and back button locations. She found that our icons clearly illustrated the desired function which eliminated the need for reading. As for the layout structure and tabs, she highlighted that icons were clear and recognizable, and buttons were bright, large, and easy to spot. She found the colour schema was consistent throughout the app and the logos stayed the same. She also noted that the information was grouped intuitively, and information stayed relevant to the page she was on. She reported that the AI chat box was unresponsive, but the help button was responsive and provided her a support email address.

4.3. Expert Evaluation **Expert Test Plan.pdf**

4.3.1. Expert Evaluation Settings

Member	Participants	Duration	Mode
Yara Albugami	Danah Alshehri	18 minutes	Remote
Danah AlOmayrini	Bassma Almanee	44 minutes	Remote
Ghala Alsaqer	Sarah Alsuwailih	35 minutes	Remote
Haya AlJadaan	Najd Abdulaziz	25 minutes	Remote

Average expert evaluation time: 30 minutes and 30 seconds

4.3.2. Expert Responses

Danah Alshehri response: Danah Alshehri Expert Test Plan

Danah Alshehri user test plan:

Danah was able to view the pop up messages that the system would display after the actions written, indicating that the visibility of the system status rule has been applied. When she tested whether there was a match between the system and the real world, she agreed that the icons were familiar. Going back between some pages was possible for her, as well as logging out, showing that users do in fact have control and freedom. She noticed consistency across the screens such as the menu bar. When she tried joining a session that was already full, it gave her a pop up message which applies the error prevention rule. The icons were clear, helping in applying the recognition rather than recall rule. A flexible and efficient to use application was achieved by having the tabs and layout support both experienced and inexperienced users. In her opinion, the aesthetic was consistent and the layout was clean, displaying relevant information. When she tried entering a wrong password, a pop up message appeared that told her what was wrong and how to fix it. She also saw that the help email was provided, supporting the help and documentation rule.

Danah Alshehri coach test plan:

Much like the previous user test plan, the system's status was visible by showing messages when actions were taken. On that note, the icons were familiar for coaches as well. She also found the user control and freedom to be applied here as well through the back and log out features. The menu bar being below and the back button being at the top showed that there was consistency across the screens. She also, similar to the user test plan, saw that the error prevention rule was applied. The icons being clear and the colors being consistent were both helpful to applying the recognition rather than recall and aesthetic and minimalist design rules, respectively. The tabs and layout both support users who have experience and users who aren't exactly experienced. If coaches were to enter an incorrect password at login, a pop up message appears. Lastly she states that the email provided helps in applying the help and documentation rule.

Basma Almanee response: Basma Almanee Expert Test Plan

Basma Almanee user test plan:

Bassma was able to understand the interface easily and found the layout clear, showing that the system supports flexibility and efficiency of use. She confirmed that the bottom menu bar worked smoothly and allowed her to move between sections without difficulty, which reflects a good match between the system and the real world. She also experienced user control and freedom, as she was able to navigate back and log out without issues. She noticed consistency across the application, especially in the placement of the menu bar and the back button. However, when evaluating visibility of system status, she mentioned that she did not test all pop-up scenarios and noticed that the system did not display users available for sending friend requests. Bassma found the icons clear and easy to recognize, supporting the recognition rather than the recorder rule. She also stated that the layout was easy to understand for different types of users. In terms of design, she mentioned that the color scheme was consistent and it was clean and simple. When testing error handling she observed that clear messages appeared when incorrect login details were entered. She also confirmed that the help feature provided an email for support, supporting the help and documentation role.

Basma Almanee coach test plan:

similar to the user test plan, best observed that the system provided feedback when actions were performed, supporting visibility of system status. She found it easy to switch between sections using the menu bar, showing a good match between the system and the real world. She confirmed that user control and freedom were present, as the logout function worked smoothly. Consistency was also maintained through the placement of navigation elements. When testing air prevention, she noted that the system played clear messages when attempting invalid actions. She also found the icons simple and easy to understand, supporting recognition rather than recall. Bassma mentioned that the layout was smooth and easy to use, supporting flexibility and efficiency of use. She also confirmed that the design was consistent and visually clean. Additionally, error messages were clear and not disruptive. Finally, she stated that the help feature was effective, as it provided clear contact information for support.

Najd Abdulaziz response: **Najd Abdulaziz Expert Test Plan**

Najd Abdulaziz user test plan:

Najd found that pop up messages weren't appearing when expected to, due to inability to send friend requests. She also found that the bottom menu bar and icons represented real-world actions and graphics were clear. To her, the back button and logout functionality worked as expected, and their placement was consistent. She was unable to join a full session due to no other users being enrolled, which didn't allow her to see the expected pop up. The icons, however, she found clear and navigable without needing to read the text due to familiarity.

She thought that the tabs indicated where actions can be taken and that the design was well-thought out in terms of usability for all users with different technological skill gaps. As for the aesthetics and design, she found the colours and logo to be consistent and that information stayed in its relevant sections without any unneeded complexity. When logging in with invalid credentials, she got a pop up message saying the user was already registered. She also found that the help and documentation rule of thumb was applied.

Najd Abdulaziz coach test plan:

When testing the visibility of system status rule, she found that pop up messages did appear when testing the given tasks. She also affirmed that our bottom menu bar icons matched real world expectations which made it recognizable. Back button and logout worked as she expected them to, and menu bar and back button locations were consistent in placement in her point of view. She also found a pop up message when trying to create a session on an invalid date, which affirmed our error prevention rule. She found that the icons used were efficient and recognizable, which made the structure easy to use without reading. As for the design layout, she found that the colours and logo were consistent and the information stayed relevant to its section. When logging in with wrong information, she found that error message pop ups were effective in error handling. When prompted with a task to check the AI Chatbox and help button, she affirmed that our help and documentation rule was fulfilled through the provided support email.

Sarah Alsuwailih response: **Sarah Alsuwailih Expert Test Plan**

Sarah Alsuwailih user test plan:

Sarah's overall user test plan findings were positive, all 10 test plans were applied. The bottom navigating tabs were easy to understand and matches the real world. Pop up messages appear immediately but it needs improvement because it is not clearly visible. Users can go to the previous page at any time. According to the consistency and standards our design is consistent and predictable. Warning messages when trying to enter a session that is full which helps in avoiding errors. Icons help users understand rather than read. The design and layout was consistent and clear. For help and diagnose and recover from errors, error messages appear when needed and they are clear and helpful. Lastly the AI chatbox and help button was available.

Sarah Alsuwailih coach test plan:

Sarah's overall coach test plan findings were positive, all 10 test plans were applied. For visibility of system status Sarah answered yes that notification appears after each action. The icons match the real world and are simple and understandable. For user control and freedom navigation was smooth and the app was responsive. Sarah answered yes that the system prevents errors by not allowing invalid inputs. Icons were easier to recognize rather than recall. Navigation was easy for all users. The design is simple, modern and consistent. Notifications of error messages are clear and help diagnose errors and recover. Lastly the AI chatbox and help button was available.

4.4. Findings

4.4.1. Overall User Evaluation Findings

A total of three participants were involved in the user evaluation, with an average evaluation duration of approximately 3.6 minutes, all conducted in person. Overall, users had a positive impression of the *Intaliq* high-fidelity prototype, particularly in terms of usability, navigation, and visual design.

Most participants described the application as simple, clean, and easy to understand, including that the interface is not overwhelming and supports quick learning. Users were able to navigate between different sections easily, especially through the bottom menu bar, which was consistently highlighted as intuitive and effective. The use of icons was also well received, as participants found them clear and helpful in understanding navigation without needing additional explanation.

In terms of design, participants appreciated the consistent color scheme, clean layout, and organized structure, which contributed to a smooth user experience. Features such as error handling during login were also effective, as users received clear and helpful feedback when entering incorrect information.

However, some usability issues were identified. One participant reported that pop-up messages for certain actions were missing, while another noted that the back button was not consistently available across all screens, which affected navigation. Furthermore, features such as the AI chatbox and help button were present but not fully functional, leading to some confusion. It was also suggested that the AI chatbox could be improved by providing example prompts to guide user interaction.

Overall, the findings indicate that while the *Intaliq* prototype demonstrates strong usability, clear navigation, and appealing design, there are some functional and consistency issues that need to be addressed to improve the overall user experience.

4.4.2. Overall Coach Evaluation Findings

Both coaches agreed on the pop up messages to be unresponsive or overall not up to standard. The bottom menu with the icons was functional for both our coaches, taking them to the desired pages accordingly. When it comes to our back button feature, Coach Yousif found it a bit frustrating that it wasn't located on every single page. On the other hand, Dr. Oluwatoyin felt the back button was available and working. Logging out of the application worked for both coaches, even showing them a pop up message for confirmation.

Coach Yousif and Dr. Oluwatoyin saw that a pop up message appeared when trying to create a session on an invalid date. However, they felt they weren't able to admit to a session that's already full since that required other users of *Intaliq* to join a session they made. To sum it up, the feature wasn't easily testable due to some circumstances. There was agreement on the fact that our icons illustrated where users were going from both coaches.

It is important to note that Coach Yousif believes the icon for the "Events" page could be better. The layout was clear for both coaches, adding that our color scheme and overall layout were good. The color schemes and logo use were consistent, as agreed on by our coaches. When they attempted to log in with an incorrect password, our coaches were presented with a pop up message that explained the error and the correct steps to take.

Just to preface the following findings, our help feature is merely supposed to have our email to contact our team in need of support. Along that, Coach Yousif saw that both the AI chatbox and help button weren't functional, due to his assumption of the help button being an actual chat between a coach and a support team member. Dr. Oluwatoyin understood that the help button was only to display the contact so she felt it worked. However she agreed that the AI chatbox didn't function for her.

Overall, they agreed that the app had a nice and clear layout, however there is definitely room for improvement when it comes to functionality.

4.4.3. Overall Expert Evaluation Findings

Analyses have concluded unanimously that both coaches and users find this system extremely usable, with excellent overall ratings resulting from our experts. All menu items were consistently located across all screens leading to all being totally confident that the criteria of consistency has been satisfied. Page designs presented in a format that closely resembles the real world and requires

little text to understand purpose, are viewed positively and provide clarity in identifying what next step to take.

Overall, from the expert responses, it was evident that the error-handling system was adequate. Typically, most participants received pop-up notifications when they attempted to join a session that's at its maximum capacity or when entering an invalid password. Therefore, they were aware of what was going on with the error they encountered while using the application. Additionally, every expert noted that the application had a very professional appearance due to its overall simplicity, which eliminated confusion for everyone. Participants also appreciated the help functions provided within the application.

There were, however, a couple major points of interest to note. First, while the pop-up notifications were effective, Sarah indicated there is still a requirement for additional work to ensure they attract the attention of the user to achieve "visibility of system status" where issues may occur. In this instance, there were conflicting reports from Najd and Basma. In particular, Najd reported that on certain occasions, the pop-up notifications did not always appear when he was attempting to send out friend requests to others. Whereas, according to Basma, there was no indication as to who could potentially become a member of her friends list. Therefore, it is clear from these examples that while the program overall is functioning well, there are minor details which could be enhanced.

4.5. Discussion

Evaluations showed that Intaliq successfully met expected criteria pertaining to usability and user experience goals. All tests conducted came back with positive feedback regarding simplicity of the application, the visual cleanness, and easy navigation. Features like the bottom bar for navigation and clear icons and layout made the interaction intuitive, familiar, and it supported usability principles like consistency and ease of use.

However, several issues were brought up by our users, coaches, and experts. Features like our pop-ups didn't always function as intended, sometimes not working or being hardly visible. The back button was noted to be missing on some screens, which limited user control. Additionally, help features like the AI chat bot and help feature were not fully implemented, which provided confusion for some users.

Overall, the design of our program was seen as user-friendly and intuitive, with some improvements needed regarding functionality, consistency, and feedback visibility. However, our application did show strong potential for changing the lack of accessible fitness communities in Saudi Arabia, which affirmed our main goal.

4.6. Conclusion

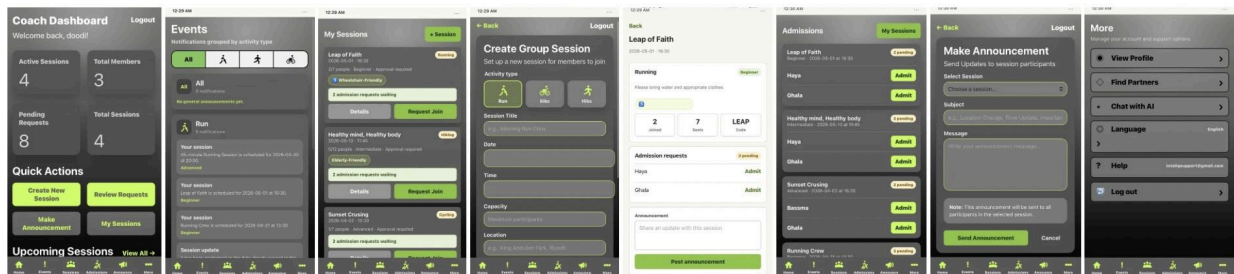
To conclude, our Intaliq prototype demonstrated strong usability, clear and consistent design, minimal visual and informational clutter, familiarity, and ease of navigation. The evaluations provided by our users, coaches, and experts affirmed that our application meets its core goals and provides a generally positive user experience.

Setting the positives aside, there was room for improvement in parts of our prototype. Particularly, this was evident in feature completion, navigation consistency, and in some cases, system feedback. Moving forward, these issues will be addressed which will enhance the program's usability and help it better meet user expectations.

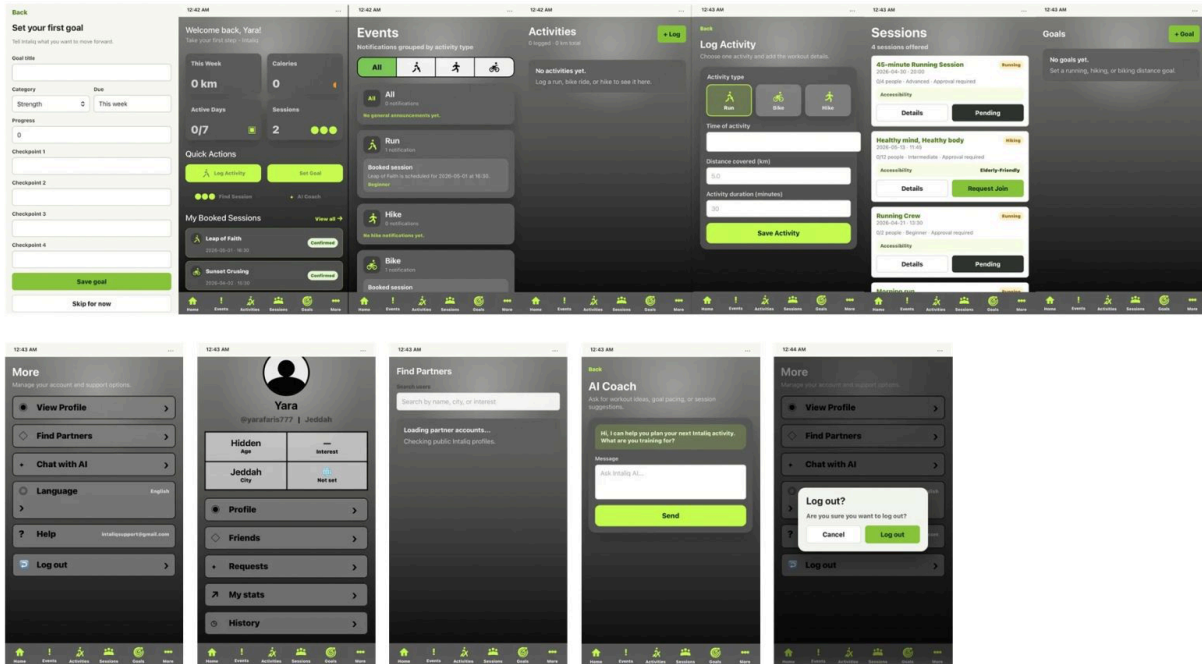
Overall, Intaliq shows strong improvement potential as a fitness platform that promotes community engagement and physical activity in one comprehensive platform, eliminating need for alternatives where information is scattered across multiple platforms, especially in Saudi Arabia.

5. Prototype Screenshots

Coach screenshots:



User screenshots:



6. References

No external references were used

7. Appendices

User test plans 📄 User Test Plans

Coach test plans 📄 Coach Test Plans

Expert test plans 📄 Expert Test Plans

Test plan templates 📄 Test plan Templates

Prototype Link.: <https://intaliq-se-365.vercel.app>

Recordings: 📄 Recordings

Video demo: 📺 Intaliq Demo.mp4

8. Individual Contribution

Member	Contribution
Yara Albugami	Interviewed a coach (Yousif), conducted user evaluation, participated in expert testing, designed and developed the high-fidelity prototype, performed final formatting and overall review of the report, wrote the high-fidelity prototype overview, including prototype Description and design improvements, wrote coach evaluation findings. Trello account: https://trello.com/invite/b/697f8f52531b1b9f96e58e75/ATTIcb1bd3a30bb47f9bca56df33e562d81cEBC74B49/se365
Haya AlJadaan	Interviewed with a coach (Dr. Oluwatoyin), conducted user evaluation, participated in expert testing, wrote the discussion and conclusion sections, organized references and appendix, and made the demo video. Trello account: https://trello.com/invite/b/697f8f52531b1b9f96e58e75/ATTI772f97c9d4197dc7eb95677d7b506d907619B00D/se365
Danah AlOmayrini	Interviewed a user (Sultanah), organized tasks amongst members, conducted user evaluation, participated in expert testing, wrote user evaluation findings, performed final formatting and overall review of the report, wrote title and introduction section. Trello account: https://trello.com/invite/b/697f8f52531b1b9f96e58e75/ATTIded601bdb32922016bcf79740daafd30E3576B75/se365
Ghala Alsaqer	Interviewed a user (Lamar), conducted user evaluation, and participated in expert testing. Trello account: https://trello.com/invite/b/697f8f52531b1b9f96e58e75/ATTI31571497c87f7f29129b3f1fa41ab00690C9BEBB/se365
Sarah Alsharif	Interviewed a user (Majed), conducted user evaluation, participated in expert testing, wrote expert evaluation findings. Trello account: https://trello.com/invite/b/6995b40f7ca01a899aecaa5c/ATTIfd364d8c5a083dd3dbd689c64e47ae17F60515D5/Sarah%20Alsharif%20Intaliq%20Progress